

Mathematics and Statistics for AI Course

Project Report

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This report discusses the use of Principal Component Analysis (PCA) for dimensionality reduction and image reconstruction.

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I. Introduction

Principal Component Analysis (PCA) is a dimensionality-reduction technique that is widely used in data science, computer vision, and machine learning. PCA was implemented in this project to a dataset of grayscale images, each changed size to 64×64 pixels and transformed into a 4096-dimensional feature vector. The goal was to mathematically analyze PCA, visualize its impact on image data, and study how the number of principal components affects reconstruction quality.

The next parts of the report clarify the conceptual bases of PCA and give a detailed discussion on the way reconstruction quality differs as the number of principal components gets bigger.

II. What Is PCA?

PCA is a linear transformation method used to decrease the dimensionality of high-dimensional data while keeping variability as possible. PCA achieves this by discovering new orthogonal axes called principal components that point in the maximum variance directions in the dataset. [1]

A. Motivation

High-dimensional datasets mostly include redundancy. For instance, images of objects that are alike may have correlated pixel intensities. PCA compresses the data by removing redundancy and projecting it onto a lower-dimensional space while still preserving most of the crucial structure.

B. Mathematical Definition

Given a dataset X with n samples (rows) and d features (columns):

1) **Compute the mean vector μ:**
   Calculated across the n samples for each feature j:
   $$\mu_j = \frac{1}{n}\sum_{i=1}^{n} X_{ij}, \quad j = 1, \ldots, d.$$
   This results in a vector $\mu \in \mathbb{R}^d$ representing the average image.
2) **Center the data**:
   Subtract the mean vector from each sample (row) to center the dataset around the origin:
   $$X_{\text{centered}} = X - \mathbf{1}_n \mu^T,$$
   where $\mathbf{1}_n \in \mathbb{R}^{n\times 1}$ is a column vector of ones.
3) **Compute the covariance matrix**:
   The covariance matrix expresses the correlation between different features (pixels):
   $$Cov[X] = \frac{1}{n-1}(X_{\text{centered}})^T (X_{\text{centered}}).$$
   This results in a symmetric matrix $\text{Cov}[X] \in \mathbb{R}^{d\times d}$.
4) **Find eigenvalues and eigenvectors**:
   Solve the characteristic equation:
   $$\text{Cov}[X]v = \lambda v,$$
   where:
   - $\text{Cov}[X]$: is the covariance matrix from step 3
   - v : is the eigenv ector, which we need to calculate
   - λ : is the eigenvalue, which we need to calculate too.

   We need to compute these values as eigenvectors v represent directions of variance, and eigenvalues λ measure how much variance lies in each direction. In python programming language, there exist a built-in libraries that compute the values of the eigenvalues and eigenvectors of any given matrix.
5) **Sort and Select Principal Components**:
   - The largest eigenvalue corresponds to the direction capturing the most variance.
   - The eigenvectors associated with the top k eigenvalues form the first k principal components.

   Let $W \in \mathbb{R}^{d\times k}$ be the matrix whose columns are these top k eigenvectors.
6) **Project the data**:
   Transform the original data into the reduced

PCA space:

$$Z = X_{\text{centered}} W,$$

where $Z \in \mathbb{R}^{n \times k}$ is the low-dimensional representation of the original data.

C. PCA in Images (Eigenfaces)

Every eigenvector can be transformed back into an image when applying PCA to image datasets. These are known as eigenfaces (or Eigenimages). They do not show real pictures, yet they represent patterns of variation like overall brightness, shapes, and edges.

D. Benefits of PCA

- Reduces computational complexity.
- Removes noise and redundancy.
- Helps visualize high-dimensional data in 2D or 3D.
- Enables efficient reconstruction and compression.

III. Does Using More Principal Components Improve Image Reconstruction?

Yes. Reconstruction quality is generally optimized when more principal components are used, as a result of keeping a bigger portion of the original variance (information) in the compressed representation. Nonetheless, this enhancement follows a pattern of diminishing returns.

A. How Reconstruction Works

Once the data is projected into a lower-dimensional PCA space, it can be reconstructed using:

$$\hat{X} = ZW^T + \mu$$

- If only a few principal components are used, the reconstruction is coarse and blurry.
- If more components are used, finer details and textures begin to appear.

B. Observations from This Project

In this assignment, reconstructions were generated using:

- 2 components
- 5 components
- 10 components
- 20 components
- 30 components

A qualitative analysis shows:

1) Using 2 Principal Components:

- Very blurry reconstruction
- Only broad shapes are visible
- Almost no texture or fine detail
- Extremely high information loss

2) Using 5 Principal Components:

- Slightly improved detail
- Coarse facial or object outlines appear
- Still far from recognizable

3) Using 10 Principal Components:

- Noticeably better
- Major structural features preserved
- Still lacks sharpness

4) Using 20 Principal Components:

- Much clearer images
- Most of the essential information is preserved
- Only fine textures are missing
- Good balance between compression and quality

5) Using 30 Principal Components:

- Very close to original images
- Fine details start appearing
- Reconstruction error significantly decreases

C. Why More Components Improve Quality

Each principal component captures a portion of total variance. The first few components capture global structures, while later components capture fine-grained details.

The cumulative variance explained increases like this (example for typical image datasets):

- First 2 components: $\sim$ 20–30% of variance
- First 10 components: $\sim$ 60–70%
- First 20 components: $\sim$ 85–90%
- First 30 components: $\sim$ 95%

Thus:

- Low component count = high compression, low quality
- High component count = low compression, high quality

D. Diminishing Returns

After a certain number of components (e.g., 30+), adding more provides only small visual improvement. Most of the meaningful structure is already captured by the first few dozen components.

IV. Trade-Off Summary

A. Principal Components Results

Principal Components	Quality	Compression Level
2	Very poor	Extremely high
5	Poor	Very high
10	Medium	High
20	Good	Moderate
30	Very good	Low
All (4096)	Perfect	None

V. Visualization

A. MSE Results by Component Count

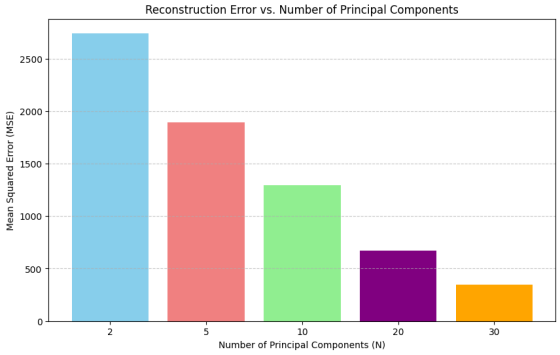


Fig. 1. Comparison of image reconstruction using 2, 5, 10, 20, and 30 Principal Components.

This analysis graph 1 explores the impact of the number of principal components (N) on the quality of image reconstruction. We calculated the Mean Squared Error (MSE) for image reconstructions using $N = 2$, $N = 5$, $N = 10$, $N = 20$, and $N = 30$ principal components.

- $N = 2$: The initial reconstruction using only 2 principal components resulted in an MSE of approximately **2742.11**. This indicates a significant loss of detail, as expected with such aggressive dimensionality reduction.
- $N = 5$: Increasing the number of principal components to 5 led to a noticeable improvement, with the MSE dropping to approximately **1894.87**. This shows that more information about the original images is retained.
- $N = 10$: Further increasing N to 10 resulted in an even lower MSE of approximately **1293.88**. The reconstruction quality continued to improve as more principal components were used to represent the data.
- $N = 20$: With 20 principal components, the MSE significantly decreased to approximately **673.92**. This demonstrates a substantial improvement in reconstruction fidelity compared to fewer components.
- $N = 30$: Using 30 principal components resulted in the lowest MSE of approximately **345.86**, indicating the best reconstruction quality among the tested values.

To Conclude this analysis, the clear trend observed is that as the number of principal components (N) increases, the Mean Squared Error (MSE) of the reconstructed images consistently decreases. This indicates that using more principal components allows for a more accurate representation of the original images, leading to higher reconstruction quality.

The trade-off lies between reducing dimensionality (for data compression or noise reduction) and maintaining sufficient detail for reconstruction. For better reconstruction quality, a higher N is generally preferred, up to a point where additional components yield diminishing returns.

B. 2D PCA Projection of Dog Images

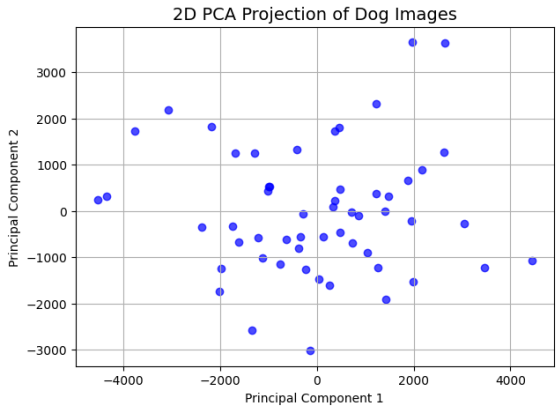


Fig. 2. Visualization of the dataset projected onto the first 2 Principal Components.

This figure 2 is a 2D PCA Projection of Dog Images. It represents the following details:

- **Points in the scatter plot:** Each blue point on the graph2 represents one of the original 55 grayscale dog images.
- **Axes:** The X-axis is labeled "Principal Component 1" and the Y-axis is labeled "Principal Component 2". These are the two most significant principal components that capture the most variance (information) from the original high-dimensional image data (4096 pixels).

By projecting the images onto these two principal components, their dimensionality was reduced from 4096 to just 2. This allows us to visualize the data and observe how the images are distributed in this reduced space. Images that are similar in terms of their features (pixels) tend to cluster together on this 2D plot 2, while dissimilar images are further apart.

In essence, this plot 2 gives a simplified view of the underlying structure and relationships within the dataset, representing any natural groupings or outliers among the dog images based on their dominant visual characteristics.

C. Top 5 Eignevalues

The top 5 eigenvalues can be displayed as follows:

- $\lambda_1 = 3.53\text{e+}06$
- $\lambda_2 = 1.89\text{e+}06$
- $\lambda_3 = 1.40\text{e+}06$
- $\lambda_4 = 1.21\text{e+}06$
- $\lambda_5 = 9.20\text{e+}05$

As previously explained, these values indicate the amount of variance captured by each principal component. The larger the eigenvalue, the more variance its corresponding principal component accounts for. The following figure 3 represents each selected eigenvalue with its corresponding principal component:

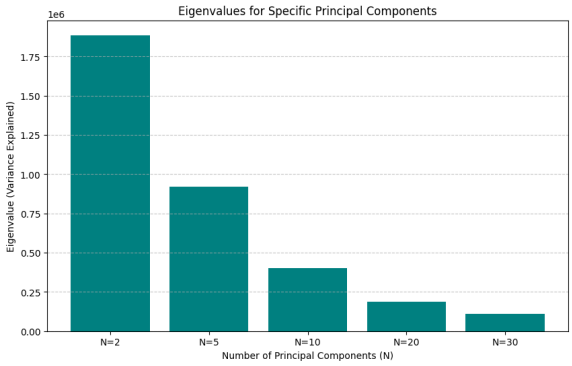


Fig. 3. Eigenvalues for Specific Principal Components

As shown in the figure 3, when the principal components increase, the eigenvalue will decrease. Thus, it is recommended to use more number of principal components.

VI. Result

When executing the code, the code will ask to upload the images that should be processed. Thus, the dogs images will be chosen and uploaded to the program. The first step applied to the pictures will be turning the images to a scale of gray color, the result of this step will result in the following figure4:



Fig. 4. Image result after applying the gray scalar.

After achieving this result, the images will be reduced and reshaped.

The following figure5 will show a comparison between the initial images after applying the gray

scalar and the result after applying the principal components to reconstruct the images:

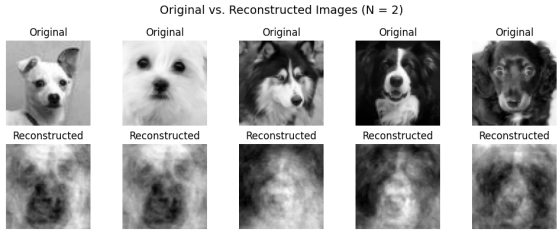


Fig. 5. Original vs. Reconstructed images.

As shown in the above figure5, the images resolution is constructed, but still preserved and can be turned into the original images.

VII. Conclusion

PCA is an effective tool for reducing dimensionality and analyzing images. It works by converting high-dimensional data into a new coordinate system specified by directions of maximum variance. PCA discloses the major patterns of variation (Eigenfaces) and permits effective image compression and reconstruction when applied to images. This project illustrated that reconstruction quality constantly enhances due to increasing the number of principal components, and that is because of maintaining more of the original information. Nevertheless, after a certain point, gains start to reduce due to diminishing returns. For the majority of image datasets, utilizing between 20 and 30 principal components results in a great balance between quality and compression efficiency. PCA remains a basic tool in computer vision and machine learning, and the use of it in this project underlines both its theoretical significance and practical success.

References

[1] S. Marukatat, "Tutorial on PCA and approximate PCA and approximate kernel PCA," *Artificial Intelligence Review*, vol. 56, no. 6, pp. 5445–5477, 2023.

[2] M. P. Deisenroth, A. A. Faisal, and C. S. Ong, *Mathematics for Machine Learning*. Cambridge, UK: Cambridge University Press, 2020.